

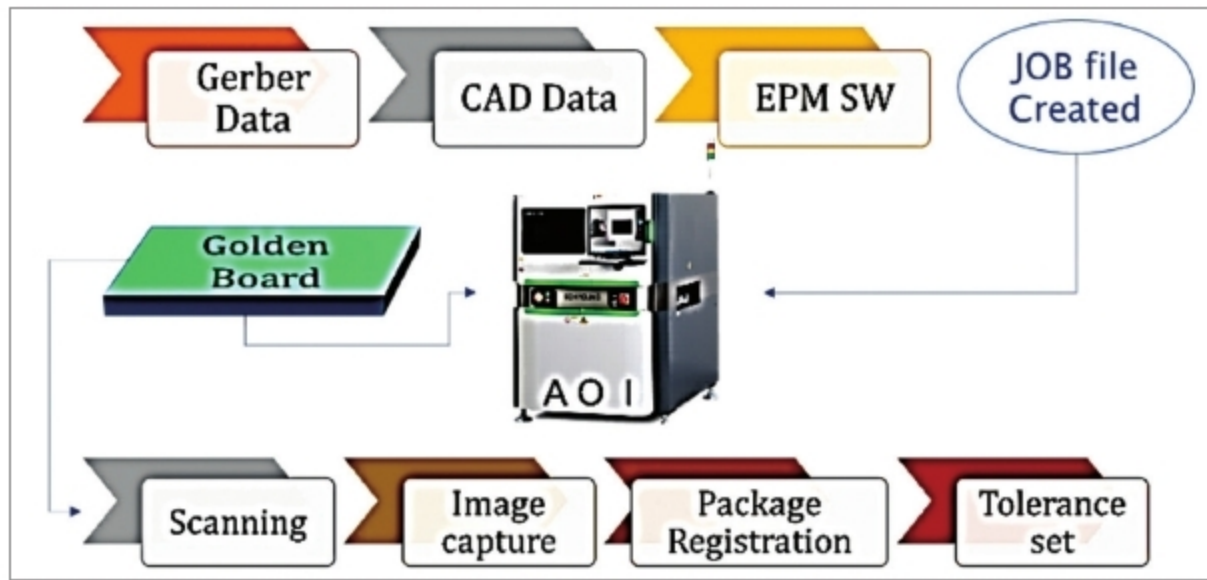


Fig. 14: AOI board inspection setup

cess is mandatory. Lots of real images must be combined with KY PCB data to get more precise information for each component.

The next step is to import the CRD

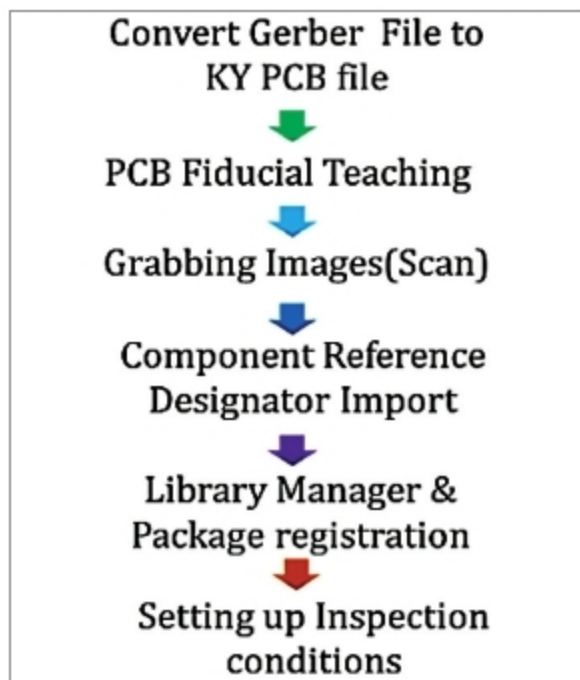


Fig. 15: Steps for creating job file (Credit: KohYoung)

file from CAD data to assign component reference designation (CRD), that is, package type, part number, x, y, and angle information.

Before setting up inspection conditions, package information (package type and size) must be registered in the library manager. This is the most important element for 3D inspection. The basic function is the separation of the mapped and unmapped components. Inspection conditions are nothing but user-defined tolerance limits so that acceptance criteria are defined with respect to specification limits.

All component information and relevant inspection conditions are saved in the job file as the final step.

3D inspection

3D inspection is nothing but an extended optical inspection system with measuring capability. Solder fillet, coplanarity, tombstone, and IC

lift are the inspection parameters that can only be measured by 3D because height measurements are difficult to detect with the 2D inspection method. High dynamic range 3D inspection technology extracts the shapes of components and solder joints and detects all types of defects.

The quality of

the solder joint can be inspected and determined from the geometric and optical properties of solder meniscus. These parameters determine the reflection properties of the meniscus formed by the solder joint. Inspection equipment measures the solder fillet (dry solder on the pad, excess solder, and insufficient solder on the pad) reflection pattern with the help of an image processing algorithm. Hence, AOI uses high-resolution cameras to detect defects in the micrometre (μm) scale.

Several optical measuring techniques are used in this process, such as:

- Laser triangulation
- Fringe pattern analysis
- Shape from shading
- Phase measuring
- Moire topography

Moire technology, fringe pattern, and phase measuring technologies are well adopted by AOI OEMs for their 3D component assembly inspection. Further, to avoid shadow and diffusion problems, they use patented technologies and solve projection obstacles by using four- or eight-way light projections for better image capture for all PCB configurations.

Summary

Compared to AOI, AXI is a newly developed inspected method. X-ray inspection makes the solder joint so direct and clear. Solder joint defects are automatically inspected through an image analysis algorithm. But AOI is a less expensive and cost-effective solution and ideal for post reflow. But in some cases, especially in RF products manufacturing, pre-reflow AOI is a widely preferred inspection process.

Currently, AOI uses fast industrial cameras that can grab up to 250 frames/sec by using 5MP CMOS sensors with remarkable image quality for high-speed applications. The improved version of AOI provides a high signal-to-noise ratio, low light performance, sharper images, and accurate colours.

AOI OEMs also consider brightness, sharpness, quantum efficiency, sensor and pixel size, resolution, dynamic range, signal to noise ratio, and contrast, which clearly differentiates between good and bad images. **EFY**

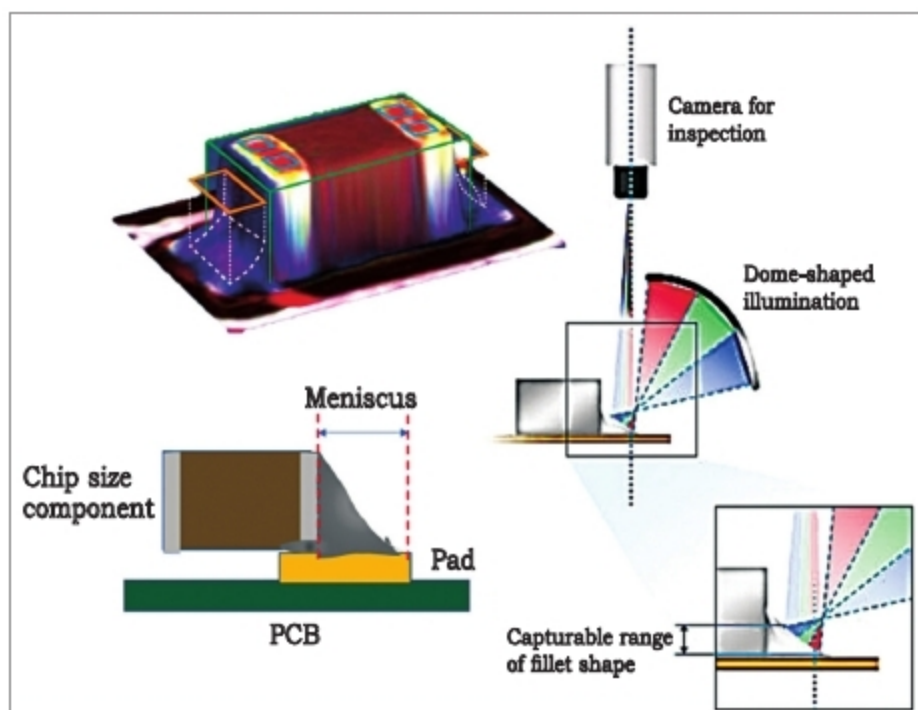


Fig. 16: 3D inspection technology extracts the shapes of components and detects defects